1 Introduction

1.1 Background

With the expansion of the unmanned vehicle (land, air, surface and underwater) in many battlefield and disaster scenarios, there is a demand for methods of controlling these. Whilst autonomy is garnering a lot of attention in this way, there is a need for the expertise and skill that a human controller brings, whilst standing them off from the dangers of hazardous environments.

An area of great interest currently is in unmanned air vehicles, where development is being taken up in earnest by organisations from major engineering firms to undergraduate project groups. Another field which is developing rapidly, however, is that of marine autonomy. This is evidenced by the speech given by the First Sea Lord at the DSEI conference, where the need for a “fleet of crewed, uncrewed, and autonomous platforms”[1] was identified. Additionally in the merchant naval space, the uncrewed platforms are being recognised, with the Workboat Code Edition 3 adding Annexe 2 in 2025 to specifically target what it terms “Remotely Operated Unmanned Vessels (ROUVs)” in section 1.1.1 [2].

Companies such as Saab AB, Thales Group, and L3Harris [3][4][5] have been making progress towards realising the vision of a “hybrid fleet” whilst companies such as ACUA Ocean, Deep Ocean Engineering, and HD Hyundai [6][7][8] are investing in ROUVs for the commercial market. Additonally, the US Navy has begun fielding ROUVs, for example the Fleet class Unmanned Surface Vessel built by Textron [9]. In all of these cases autonomy has been the end goal, to perform dangerous task such as mine countermeasures, tedious ones such as survey work [10], and other routine but challenging tasks such as docking [11].

Autonomy in surface ships is a significant challenge, however. Any oceangoing vessel must comply with national and international law, which not only set requirements for the vessels but also for the behaviours that they must exhibit. A key example of this is the Convention on the International Regulations for Preventing Collisions at Sea (COLREGs) [12]. COLREGs set out rules determining the lights, shapes, and sounds that a vessel must display depending on its state, but also they describe responsibilities of different vessels in different situations. These rules are often subjective, however, and were designed for human navigators [11]. Implementing COLREGs into autonomous systems is as such an area of active development.

To operate in national waters, a ship must be approved by a national agency, which in the UK is the Maritime and Coastguard Agency (MCA). To do this, the vessel must show that it is compliant with Workboat Code Edition 3 (WBC3). ROUVs are now goverened specifically by Annexe 2, which sets out requirements on the ROUVs and their control stations. The control stations on a ROUV can be varied, from a wheelhouse on a retrofit vessel to a land based operations centre. Of note is that ROUVs through WBC3 must comply with a number of standards [13], including BS EN 61508: 2010 [14]. This standards concerns the functional safety of electrical/electronic/programmable electronic safety related systems, such as the emergency stop system on a vessel or its damage control systems. BS EN 61508 defines Safety Integrity Levels (SIL), which correspond to a number of dangerous failurs per hour. For example, achieving SIL 1 compliance requires a failure rate no greater than 10^{-5} dangerous failures per hour, whilst SIL 3 requires no greater than 10^{-9}. Whilst implementations of BS EN 61508 are common in many sectors (for example, ISO 26262 for the automotive industry [15]), ensuring that any automated controllers are also compliant is a challenge. To ensure that an ROUV has a safety case sufficient to be granted approval from the MCA, remote crew operation capability is often required.

1.2 Problem Statement

As mentioned above, autonomy in workboats is as such becoming increasingly mature and is being deployed to many different platforms. Until a level of assurance in the autonomous controller’s capabilies and reliability is reached (if such a level will ever be reached), human oversight of the vessel will be required in some from. Even in situations where autonomy is not the end goal for the vessel, there is a need for a method of removing the crew from the vessel in hazardous conditions. This can reduce the chance and/or severity of accidents (man overboard, injuries from machinery, loss of platform) whilst potentially improving quality of life for those crews (e.g. protection from the elements). Moreover, there is an opportunity for workboats to be constructed without a wheelhouse and on board controls, which could reduce the cost, complexity, and displacement of the platform.

In all of these situations remote control of the vessel is desirable: to keep watch over autonomy or remove the crew from the vessel. As mentioned above, Workboat Code edition 3 Annexe 2 [2] prescribes requirements for an ROUV, but does not prescribe solutions to this issue.

Existing control stations for ROUVs are large and bulky. They often are entire rooms, with multiple monitors, a PC like interface, and additional hardware controls set up with communications software built in to enable control or keeping watch of the ROUV beyond visual range. In the case where they are designed for use outside (such as in the case of line of sight control or), they are still too large and heavy to be carried by an operator, needing instead to be set up from a single vantage point (such as the Deep Trekker BRIDGE Console [16] and MarineNav Oceanus [17]). This is ideal for situations that are beyond visual range, such as mission operation, however within line of sight this solution becomes significantly less desirable, as being able to walk around a harbour or be aboard a vessel whilst controlling it may provide better situational awareness. A possible solution is the SRoC Series of ground control station provided by UXV Technologies [18]. It is a control tablet that is man portable, providing control of a remote vehicle over radio communications using hardware buttons and joysticks built into the tablet. This solution is however restricted in its platform agnostic design, failing to offer specific needs of a marine craft (such as throttle control) and integrates with a proprietary communications suite.

As such, there is a need of a bespoke, marine centred remote control unit (RCU) which is man portable and capable of providing remote control of an ROUV within line of sight.

1.3 System Narrative

In designing an RCU, the manner in which the operator issues commands and receives information is important. The RCU will have to be provide situational awareness that is at least as good as a crew member would have aboard the vessel. It also must be able to provide the same functionality and level of control as those controls normally found aboard the platform. Additionally, it must be capable of connecting to the vessel in a reliable and secure way. Such a communication channel must be capable of providing an appropriate SIL, and must permit the transmission of commands, information, and alerts simultaneously. There a number of approaches to this that are dependant on the setting in which the controller would be used.

There are also specific needs that an RCU will need to meet introduced by the remote control. The platform can only accept one controller at a time to avoid complexity in controller hierarchy. As such, the RCU will need to manage assuming control of the vessel, both from a "cold start" and from an exant controller. Additionally, the RCU will need to manage different methods of control behaviour (open water operation, confined spaces, and dynamic station keeping) whilst also being able to manage a degraded communication environment (such as in inclement weather).

Managing ROUV subsystems is also core functionality that must be met. Control of the propulsion and power systems is necessary for controlling movement and for enabling a "cold start" scenario. COLREGs 1972 mandates shapes, lights, and sounds that must be managed depending on the boat's situation and movements, and must be controlled as well [12]. Additionally, the emergency system aboard the vessel must be controlled as well. This includes both alerts to the operator and the emergency stop functionalities. To that end, the vessel must also comply with all relevent legislation, and the RCU must enable the vessel to meet those requirements [2] [12].

1.4 Aims and Objectives

This report aims to provide an initial architectural solution to a man portable RCU that can control an ROUV within line of sight in compliance with all relevant legislation, which can be then used to inform the design of a realised solution.

To do this, this report has the following objectives:

1. To identify scenarios, capabilites, and use cases for an RCU, whilst understanding the relationships between different use cases and the actors external to the system.

2. To define a set of user activities that the RCU would need to be capable of facilitating derived from the scnearios and use cases.

3. To derive a set of requirements from the activities, use cases, and relevant legislation, and to functionally decompose them to inform the structure of the RCU.

4. To identify the constituent parts of a solution, defining their properties and internal structures, allocating them to the functional requirements, and structuring them such that the RCU would be able to facilitate the user activities and scenarios.
5. To supplement the behavioural model of the architected solution with respect to the structral model, to further understand the interaction between the actors and subsystems.

2 Technical Results

2.1 Functional Requirements Definition

2.2 Functional Decomposition, Elaboration, and Allocation

2.3 System Structure Definition

2.4 System Interface Behaviour

2.5 Alternative Behaviours

2.6 Architecture Analysis

3 Conclusion

3.1 System Specification

3.2 Achievement of Aims and Objectives

3.3 Highlights and Recommendations

References

[1] G. Jenkins, *First sea lord general sir gwyn jenkins speech at dsei 2025*, Sep. 2025. [Online]. Available: https://www.gov.uk/government/speeches/first-sea-lord-general-sir-gwyn-jenkins-speech-at-dsei-2025.

[2] Ship Standards, "Workboat code edition 3," Maritime and Coastguard Agency, Tech. Rep., Jan. 2025. [Online]. Available: www.gov.uk/government/organisations/maritime-and-coastguard-agency.

[3] Saab AB. "Naval autonomy | naval | saab. "[Online]. Available: www.saab.com/products/naval/naval-autonomy.

[4] Thales Group. "Naval drone warfare | thales group. "[Online]. Available: https://www.thalesgroup.com/en/solutions-catalogue/defence/naval/naval-drone-warfare.

[5] L3Harris. "Autonomous systems | l3harris® fast. forward. "[Online]. Available: https://www.l3harris.com/all-capabilities/autonomous-systems.

[6] ACUA Ocean. "Home | acua ocean | uncrewed surface vessel | usv | plymouth, uk. "[Online]. Available: www.ocean.tech.

[7] Deep Ocean. "Underwater rov, usv - underwater drones - deep ocean engineering. "[Online]. Available: www.deepocean.com.

[8] HD Hyundai Co. Ltd. "Avikus | hd hyundai. "[Online]. Available: www.hd.com/en/business/shipbuilding/avikus/contents.

[9] Textron Systems. "Common unmanned surface vehicle | textron systems. "[Online]. Available: https://www.textronsystems.com/products/cusv.

[10] J. Manley, "Oceans 2008," in *Unmanned Surface Vehicles, 15 Years of Development*, IEEE, 2008, ISBN: 9781424426201. [Online]. Available: https://www.scopus.com/pages/publications/70350103995?origin=resultslist.

[11] S. Campbell, W. Naeem, and G. W. Irwin, "A review on improving the autonomy of unmanned surface vehicles through intelligent collision avoidance manoeuvres," *Annual Reviews in Control*, vol. 36, pp. 267–283, 2 Sep. 2012, ISSN: 13675788. DOI: 10.1016/j.arcontrol.2012.09.008. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1367578812000430?getft_integrator=scopus&pes=vor#s0030.

[12] International Maritime Organization, "1972 convention on the international regulations for preventing collisions at sea," International Maritime Organization, Tech. Rep., Oct. 1972. [Online]. Available: www.cil.nus.edu.sg.

[13] Ship Standards, "Marine information note min 698 amendment 1 (m) workboat code edition 3-standards and guidelines for best practice," Maritime and Coastguard Agency, Tech. Rep., Dec. 2024. [Online]. Available: https://assets.publishing.service.gov.uk/media/67654c6fcdb5e64b69e30912/MIN_698_Amendment_1.pdf.

[14] IEC, "Bs en 61508-1:2010 functional safety of electrical/electronic/programmable electronic safety-related-systems - part 1: General requirements," British Standards Institute, Tech. Rep., Jun. 2010.

[15] ISO, "Bs iso 26262-1:2018 road vehicles - functional safety part 1: Vocabulary," British Standards Institute, Tech. Rep., Dec. 2018.

[16] D. Trekker. "Bridge console | advanced rov control station | deep trekker. "[Online]. Available: https://www.deeptrekker.com/shop/products/deep-trekker-display-console.

[17] MARINENAV. “Oceanus ultimate rov system. ”[Online]. Available: https://www.marinenav.ca/oceanus-ultimate-rov.html.

[18] U. Technologies. “Sroc 7" linux - enterprise ground control station (gcs). ”[Online]. Available: https://www.uxvtechnologies.com/ground-control-stations/sroc-linux-7.